POLYNOMIAL BICOMODULES ARE PARAMETRIC RIGHT ADJOINTS (AFTER SPIVAK–GARNER–FAIRBANKS), WITH THE WORKER/HARNESS WELD AS A WORKED INSTANCE

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Abstract. Spivak, Garner and Fairbanks proved that, for polynomial comonoids (equivalently small categories) c and d, the category of (c, d)-bicomodules in **Poly** is equivalent to the category of parametric right adjoint functors c-**Set** $\to$ d-**Set**. We restate this correspondence precisely, with its slice (aggregation) factorization, and then run it on a single concrete bicomodule drawn from the semantics of stateful agents: the worker/harness *weld* $B = \Delta S \triangleleft \Delta T$ of two store comonads. We prove that the weld is the *regular* $(\Delta T, \Delta S)$-bicomodule and that it represents the prafunctor $\bar{F}_B(V) = (T \times V)^S = T^S \cdot V^S$; its slice factorization keeps the power $(-)^S$ as connected-limit content and aggregates over the branch-map set T^S by a coproduct Σ. We then locate the *worker* $A = \Delta(S \times T)$ inside this picture: read as a $(\Delta T, \Delta S)$-bicomodule of the same variance as B, it is exactly the constant-branch sub-bicomodule $B' \subseteq B$, so worker and weld *share* the content power $(-)^S$ and differ only in their Σ-index, T versus T^S. This decides a former conjecture of ours ("worker = parametric right adjoint"): affirmatively but with a sharp correction — the worker is not the whole weld-prafunctor but its constant-branch sub-bicomodule, with proper inclusion exactly when $|S| \geq 2$ and $|T| \geq 2$.

1. Introduction

This note has two jobs. The first is expository: to state, carefully and with correct attribution, the theorem of Spivak, Garner and Fairbanks [4] identifying polynomial bicomodules with parametric right adjoints. The second is a worked instance: to run that theorem on one concrete bicomodule arising in the semantics of stateful, interacting agents, and in doing so to settle a question we had earlier been able only to conjecture.

The correspondence. In the polynomial-functor framework [3], a small category is the same datum as a polynomial comonoid in $(\mathbf{Poly}, \triangleleft, \mathsf{y})$ [1]. The "horizontal" maps between two such categories c, d are the two-sided *bicomodules*, and Spivak, Garner and Fairbanks [4] identify them concretely: a (c, d)-bicomodule is exactly a *parametric right adjoint* functor c-**Set** $\to$ d-**Set** between the associated copresheaf categories. This is their Proposition 3.9(5) together with Theorem 4.28. Moreover every prafunctor factors (Definition 3.5, equation (21)) through a slice, the first factor carrying the genuine right-adjoint (connected-limit) content and a coproduct Σ aggregating a tail that is not itself bicomodule data.

A note on attribution. The result is sometimes loosely attributed to "Garner 2019." That attribution is not citable: it refers to a HoTTEST *talk* (R. Garner, 11 December 2019), an announcement, not a written proof. The definitive, peer-reviewed written source is Spivak, Garner and Fairbanks, *Functorial Aggregation*, J. Pure Appl. Algebra **229**(2) (2025), art. 107883; arXiv:2111.10968 [4]. Garner is credited there, as a co-author of the actual paper. We cite [4] throughout and do not treat the 2019 talk as a proof.

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[0]Supersedes the 2026-09-20 draft; corrects attribution and resolves the former Conjecture 5.1.

The worked instance: the weld. In models of stateful agents one carries a piece of local state S by the *store comonad* $\Delta S = S \cdot \mathsf{y}^S$. Two agents with states S and T can be welded by functor composition, $B = \Delta S \triangleleft \Delta T$ — the "store of stores" used by Braithwaite–Hedges–Mihejevs [5]. A competing discipline entangles the states by the Dirichlet product into a single store, $A = \Delta S \otimes \Delta T = \Delta(S \times T)$, which we call the *worker*. We earlier conjectured (in the superseded 2026-09-20 draft) that the worker "is" the parametric right adjoint content of the weld, with the weld's extra data as the Σ-aggregation tail. We can now say precisely what is true.

The weld B *is* a bicomodule — the regular $(\Delta T, \Delta S)$-bicomodule (§4) — and it represents the prafunctor $\bar{F}_B(V) = (T \times V)^S$, whose factorization keeps the power $(-)^S$ as content and aggregates over the branch-map set T^S. The worker A, read consistently as a bicomodule of the same variance, is the *constant-branch sub-bicomodule* $B' \subseteq B$ (§5). Worker and weld share the content power $(-)^S$; they differ only in their Σ-index, T versus T^S. So the former conjecture is settled affirmatively in a precise and corrected form: the worker is a bicomodule, and it is not the whole weld-prafunctor but its constant-branch sub-bicomodule.

Status of the claims. The correspondence of §3 is [4]'s established theorem; we state it and sketch its proof in our own words, and claim no part of it. The weld facts of §§4–5 are our own results; the regular-bicomodule structure and the represented prafunctor $\bar{F}_B(V) = (T \times V)^S$ have additionally been peer-reviewed by a referee (Rick), and the worker-as-constant-branch-slice comparison is proved and machine-checked at small shapes. Trust grades are flagged at each statement.

Outline. Section 2 fixes polynomial conventions, the store comonad, parametric right adjoints, and the bicomodule definition. Section 3 states the SGF correspondence, its slice factorization, and the handedness convention. Section 4 proves the weld is the regular bicomodule and computes its prafunctor. Section 5 locates the worker as the constant-branch sub-bicomodule and decides the former conjecture. Section 6 draws the consequence for the compositional-correctness program.

2. Preliminaries

We work in the category **Poly** of polynomial functors **Set** $\to$ **Set**, following the conventions of [3].

Definition 2.1 (polynomials and their maps)**.** A *polynomial* is a coproduct of representables $p = \sum_{s \in S_p} \mathsf{y}^{p[s]}$, acting on sets by $p(X) = \sum_{s \in S_p} X^{p[s]}$; we call S_p its *positions* (or shapes) and $p[s]$ its set of *directions* at s. A morphism $f\colon p \to q$ is a pair $f = (f_1, f^\sharp)$ with $f_1\colon S_p \to S_q$ forward on positions and $f^\sharp_s\colon q[f_1 s] \to p[s]$ *backward* on directions. Composition is $(g \circ f)_1 = g_1 f_1$ and $(g \circ f)^\sharp_s = f^\sharp_s \circ g^\sharp_{f_1 s}$: backward maps compose in reverse. Under the full embedding **Poly** $\hookrightarrow [\mathbf{Set}, \mathbf{Set}]$ a morphism of polynomials *is* a natural transformation.

Poly carries two monoidal structures we need.

Definition 2.2 (the two products)**.** The *composition product* is $(p \triangleleft q)(X) = p(q(X))$, with unit y; it is strictly associative. The *Dirichlet product* $p \otimes q$ has positions $S_p \times S_q$ and directions $p[s] \times q[t]$ at (s, t), with unit y.

Definition 2.3 (polynomial comonoids are categories)**.** A comonoid for $(\mathbf{Poly}, \triangleleft, \mathsf{y})$ is the same data as a small category [1, 3]: positions are objects, directions at s are the morphisms out of s, the counit ε picks out identities and the comultiplication δ composes. The framed bicategory of polynomial comonoids is written $\mathbf{Cat}^\sharp$ [3]. A polynomial comonoid c presents a category we also call c, with copresheaf category c-**Set**.

Definition 2.4 (store comonad)**.** For a set S, the *store* (or costate) polynomial is $\Delta S := S \cdot \mathsf{y}^S$ — positions S, with $\Delta S[s] = S$ at every position — carrying the store comonad $\Delta S(X) = S \times X^S$. Its comultiplication is $\delta_S(s, \varphi) = (s,\, i \mapsto (i, \varphi))$ and counit $\varepsilon_S(s, \varphi) = \varphi(s)$. As a category ΔS is the *indiscrete* (codiscrete) category on S: exactly one morphism between any two objects. For $S \neq \emptyset$ it is equivalent to the terminal category 1, so ΔS-**Set** $\simeq$ **Set**.

The two objects of study on the SGF side are parametric right adjoints and bicomodules.

Definition 2.5 (parametric right adjoint)**.** Let C have a terminal object 1. A functor $F\colon C \to D$ is a *parametric right adjoint* (PRA), or *prafunctor*, if the first factor of the canonical slice factorization

$$C \xrightarrow{F/1} D/F1 \xrightarrow{U} D$$

is a right adjoint, where $(F/1)(c) = (Fc \to F1)$ and U forgets the slice down to D [4, Def. 3.5, eq. (21)]. For endofunctors of **Set** this is equivalent to preserving connected limits, equivalently to being polynomial [4, Prop. 2.5].

Definition 2.6 (bicomodule)**.** Let c, d be polynomial comonoids. A (c, d)*-bicomodule* [4, Def. 4.2] is a polynomial m with a left d-coaction $\lambda\colon m \to d \triangleleft m$ and a right c-coaction $\rho\colon m \to m \triangleleft c$,

$$d \triangleleft m \xleftarrow{\lambda} m \xrightarrow{\rho} m \triangleleft c,$$

satisfying the left comodule laws (counit and coassociativity, [4, (34)]), the right comodule laws ([4, (35)]), and the compatibility law $(\lambda \triangleleft c) \circ \rho = (d \triangleleft \rho) \circ \lambda$ ([4, (36)]). Bicomodules between c and d are the "horizontal" (loose) maps of $\mathbf{Cat}^\sharp$.

3. The Spivak–Garner–Fairbanks correspondence

We restate the results of [4] this note leans on, in our own words. All are established; we claim no part of them.

Theorem 3.1 ([4, Thm. 4.28])**.** *The framed bicategories* $\mathbf{Cat}^\sharp$ *and* Comod(**Poly**) *are equivalent. Under the equivalence: objects are small categories, i.e. polynomial comonoids; the horizontal (loose) maps from c to d are the (c, d)-bicomodules; the vertical (tight) maps are the comonoid homomorphisms, i.e. retrofunctors (*not *ordinary functors); and the 2-cells are bicomodule morphisms.*

The pointwise identification of the horizontal maps is a six-way equivalence, of which the piece we use is item (5): parametric right adjoints.

Proposition 3.2 ([4, Prop. 3.9(5), with Thm. 4.28])**.** *For polynomial comonoids c, d there is an equivalence of categories*

$$\mathrm{Comod}(\mathbf{Poly})(c, d) \;\simeq\; \mathrm{PRA}\big(c\text{-}\mathbf{Set},\, d\text{-}\mathbf{Set}\big) \;=\; \mathrm{CLP}\big(c\text{-}\mathbf{Set},\, d\text{-}\mathbf{Set}\big),$$

where CLP *denotes the connected-limit-preserving functors. That is, a (c, d)-bicomodule* is *a parametric right adjoint functor c-**Set** $\to$ d-**Set**. (Prop. 3.9 lists six equivalent descriptions of the hom-category, with the bicomodule description supplied via Theorem 3.1; items also include functors $d \to \mathbf{Set}[c]$ and $d \to \mathrm{Fam}((c\text{-}\mathbf{Set})^{\mathrm{op}})$.)*

Proof sketch, following [4]. A (c, d)-bicomodule m determines a functor c-**Set** $\to$ d-**Set** by right Kan extension / coend against its carrier, using the two coactions to make the output a d-copresheaf and to integrate over c. That this functor is a parametric right adjoint is the content of [4, §2–§3]: polynomial endofunctors of **Set** are exactly the connected-limit-preserving ones ([4, Prop. 2.5]), and the dependent-sum indexing of a bicomodule carrier realises the slice factorization of Definition 2.5. Conversely a prafunctor is reconstructed from the slice data $F/1$ and the indexing set $F1$, which

assemble into a bicomodule carrier with its coactions; composition is covariant and matches ordinary functor composition via the coreflexive-equalizer composite $\odot_d$, **Poly** preserving coreflexive equalizers ([4, Lem. 2.19]). The two passages are mutually inverse up to coherent isomorphism. □

The factorization we actually compare against is the slice factorization of a single prafunctor, because it is what performs the "content versus aggregation" split.

Proposition 3.3 (slice factorization, [4, Def. 3.5, eq. (21)])**.** *Every prafunctor* $F\colon c\text{-}\mathbf{Set} \to \mathbf{Set}$ *factors as*

$$c\text{-}\mathbf{Set} \xrightarrow{F/1} \mathbf{Set}/F1 \xrightarrow{\Sigma} \mathbf{Set},$$

where $F/1$ *is a* right adjoint *— the genuine polynomial/bicomodule content — and* Σ *is the coproduct over the index set* $F1$*, a* left adjoint *"aggregation tail" that carries no bicomodule data. (The factorization is the datum named in Definition 3.5, equation (21); it is elaborated inside the proof of* [4, Prop. 3.6]*, which is itself the single-variable equivalence statement.)*

The moral is a content/aggregation split: the bicomodule content lives in the right-adjoint factor $F/1$, and the coproduct Σ is a tail the correspondence does not see as bicomodule data. A second, equivalent presentation of a prafunctor is the data-migration form, which we record because it names the pieces the weld will exhibit.

Proposition 3.4 (data-migration form, [4, Prop. 3.20])**.** *Every prafunctor factors as* $\Sigma_T \circ \Pi_\pi \circ \Delta_S$*: a reindexing* Δ_S *(pullback), a dependent product* Π_π *(the connected-limit content), and a dependent sum* Σ_T *(the aggregation tail).*

Remark 3.5 (handedness)**.** We use [4]'s convention throughout: a (c, d)-bicomodule $m\colon c \to d$ corresponds to a parametric right adjoint $c\text{-}\mathbf{Set} \to d\text{-}\mathbf{Set}$ (Proposition 3.2). The reader should be warned that Lynch, Shapiro and Spivak, *All concepts are* $\mathbf{Cat}^\sharp$ [2], adopt the *reversed* handedness, sending a bicomodule to a functor $d\text{-}\mathbf{Set} \to c\text{-}\mathbf{Set}$. It is the same theorem under the opposite variance on bicomodules; any concrete identification must pin the convention first. We pin [4]'s.

4. The weld is the regular bicomodule

We now run the correspondence on the concrete bicomodule of interest. Fix nonempty sets S, T and the store comonoids $\Delta S, \Delta T$ of Definition 2.4. The *weld* is the composition product

$$B \;:=\; \Delta S \triangleleft \Delta T,$$

with positions $\{(s, g) : s \in S,\ g\colon S \to T\} = S \times T^S$ and directions $S \times T$ everywhere, so a position of B carries a whole *branch map* $g\colon S \to T$.

Lemma 4.1 (regular bicomodule)**.** *In any monoidal category* $(\mathcal{V}, \triangleleft, \mathsf{y})$*, for comonoids* $c = (c, \varepsilon_c, \delta_c)$ *and* $d = (d, \varepsilon_d, \delta_d)$*, the object* $d \triangleleft c$ *is a* (c, d)*-bicomodule with*

$$\lambda := \delta_d \triangleleft c\colon d \triangleleft c \to d \triangleleft d \triangleleft c = d \triangleleft (d \triangleleft c), \qquad \rho := d \triangleleft \delta_c\colon d \triangleleft c \to d \triangleleft c \triangleleft c = (d \triangleleft c) \triangleleft c.$$

Proof. The left comodule laws [4, (34)] are the comonoid laws of d — $(\varepsilon_d \triangleleft d) \circ \delta_d = \mathrm{id}$ and $(\delta_d \triangleleft d) \circ \delta_d = (d \triangleleft \delta_d) \circ \delta_d$ — whiskered on the right by $\triangleleft c$; the right laws [4, (35)] are the comonoid laws of c whiskered on the left by $d \triangleleft$. For compatibility [4, (36)], λ acts on the first (d) factor and ρ on the last (c) factor; since $\triangleleft$ is a bifunctor, the two whiskerings commute by the interchange law, and both $(\lambda \triangleleft c) \circ \rho$ and $(d \triangleleft \rho) \circ \lambda$ equal the single map $\delta_d \triangleleft \delta_c\colon d \triangleleft c \to d \triangleleft d \triangleleft c \triangleleft c$. □

Theorem 4.2 (weld as regular bicomodule)**.** *The weld* $B = \Delta S \triangleleft \Delta T$ *is the regular* $(\Delta T, \Delta S)$*-bicomodule: take* $c = \Delta T$*,* $d = \Delta S$*, carrier* $d \triangleleft c = \Delta S \triangleleft \Delta T = B$*, with* $\lambda = \delta_{\Delta S} \triangleleft \Delta T$ *and* $\rho = \Delta S \triangleleft \delta_{\Delta T}$*. Hence* B *is a horizontal map of* $\mathbf{Cat}^\sharp$*, i.e. a prafunctor* $\Delta T\text{-}\mathbf{Set} \to \Delta S\text{-}\mathbf{Set}$*.*

Proof. Immediate from Lemma 4.1 with $c = \Delta T$, $d = \Delta S$, since $\Delta S, \Delta T$ are comonoids (Definition 2.4). The pair (c, d) in the sense of Proposition 3.2 is $(\Delta T, \Delta S)$. □

Remark 4.3 (trust)**.** Theorem 4.2 is *proved* and *peer-reviewed* (referee: Rick, clause (1)/Thm 2.2). All five bicomodule axioms were additionally machine-checked as equalities of natural transformations for $(|S|,|T|) \in \{(2,2),(2,3),(3,2),(1,2),(2,1)\}$. Lemma 4.1 is standard category theory (a $\triangleleft$-product of comonoids is the regular bicomodule between them); the content here is its use to pin the pair $(\Delta T, \Delta S)$ for the weld, with no ad hoc choice required.

The theorem removes the only genuine type-level worry about the weld: although $\Delta S \triangleleft \Delta T$ is a priori a *product* of two comonoids rather than a map *between* two fixed ones, the regular-bicomodule construction turns the $\triangleleft$-product of any two comonoids into a bicomodule between them.

The represented prafunctor. Because ΔS is the indiscrete category on S, it is equivalent to the terminal category, so ΔS-**Set** $\simeq$ **Set** (Definition 2.4); likewise for ΔT. Right-adjointness and the slice factorization are equivalence-invariant, so we transport the weld prafunctor to an endofunctor $\bar{F}_B$ of **Set**.

Theorem 4.4 (represented prafunctor)**.** $\bar{F}_B(V) \cong (T \times V)^S = T^S \times V^S = \sum_{g \in T^S} V^S$. *As a polynomial endofunctor of* **Set**, $\bar{F}_B = T^S \cdot \mathsf{y}^S$*: positions* T^S*, directions* S*. Concretely* $\bar{F}_B(X) = \{(g,h) : g\colon S \to T,\ h\colon S \to X\}$.

Proof. The regular bicomodule $d \triangleleft c$ is the composite, over the unit y, of two one-sided comodules,

$$\Delta T \xrightarrow{G} \mathsf{y} \xrightarrow{H} \Delta S, \qquad \text{carrier } H \triangleleft G = \Delta S \triangleleft \Delta T = B,$$

where G is the right ΔT-comodule with carrier ΔT and $\rho = \delta_{\Delta T}$, and H is the left ΔS-comodule with carrier ΔS and $\lambda = \delta_{\Delta S}$. Bicomodule composition over the unit y is just $\triangleleft$ (no equalizer is needed), so prafunctors compose and $\bar{F}_B = \bar{F}_H \circ \bar{F}_G$.

The factor $\bar{F}_G\colon \Delta T$-**Set** $\to$ **Set**. As a ΔT-variable polynomial (the **Set**$[c]$ description of [4, Prop. 3.9]) the right comodule with carrier ΔT is $F_G(X) = \sum_{t \in T} \Delta T\text{-}\mathbf{Set}(\mathsf{y}_t, X) = \sum_{t \in T} X(t)$, where y_t is the representable at t. In the indiscrete category y_t is the terminal copresheaf, so ΔT-**Set**$(\mathsf{y}_t, X) = \lim X$; hence $F_G(X) = \sum_{t \in T} \lim X$, and after transport $\bar{F}_G(V) = T \times V$.

The factor $\bar{F}_H\colon$ **Set** $\to \Delta S$-**Set**. A left ΔS-comodule is a functor $\Delta S \to$ **Poly** (the Prop. 5.7 dictionary of [4]); the regular one (carrier ΔS, $\lambda = \delta_{\Delta S}$) assigns to each object $a \in S$ the component polynomial y^S (read off δ_S: one position over a, with S directions). So $F_H(W)(a) = (\mathsf{y}^S)(W) = W^S$, and after transport $\bar{F}_H(W) = W^S$.

Therefore $\bar{F}_B(V) = \bar{F}_H(\bar{F}_G(V)) = (T \times V)^S = T^S \times V^S$, and $(T \times V)^S = \{S \to T \times V\} = \{(g,h) : g\colon S \to T,\ h\colon S \to V\}$. □

Remark 4.5 (trust, and the point of the computation)**.** Theorem 4.4 is *proved* and *peer-reviewed* (Rick re-derived $\bar{F}_B$ on seven shapes). Note that $\bar{F}_B = T^S\mathsf{y}^S$ is *neither* the underlying polynomial of the worker $A = (S \times T)\mathsf{y}^{S \times T}$ *nor* that of the carrier $B = (S \cdot T^S)\mathsf{y}^{S \times T}$: the prafunctor represented by a bicomodule is not the underlying polynomial functor of its carrier. (Cardinalities at $|S| = |T| = 2$, $V = 2$: $\bar{F}_B = 16$, A-underlying $= 64$, B-underlying $= 128$.)

Applying Proposition 3.3 to $\bar{F}_B(V) = \sum_{g \in T^S} V^S$:

$$\bar{F}_B(1) = T^S, \qquad (\bar{F}_B/1)(V) = \left(V^S\right)_{g \in T^S} = \Delta_{T^S} \circ (-)^S, \qquad \Sigma\colon \mathbf{Set}/T^S \to \mathbf{Set}.$$

The content factor $\bar{F}_B/1$ is a right adjoint — the power $(-)^S$ (right adjoint to the copower $S \times (-)$) followed by the diagonal Δ_{T^S} (right adjoint to Σ). The connected-limit content is the **power** $(-)^S$; the aggregation tail is the coproduct Σ over the **branch-map set** $T^S = \mathrm{Hom}(S,T)$.

5. The worker is the constant-branch sub-bicomodule

The *worker* is the Dirichlet entanglement

$$A \;:=\; \Delta S \otimes \Delta T \;=\; \Delta(S \times T),$$

with positions $S \times T$ and directions $S \times T$ everywhere. A priori A is a 0-*cell* (a comonoid/category), whereas the weld content is a 1-*cell* ($\Delta T \to \Delta S$); to compare them we must read A as a $(\Delta T, \Delta S)$-bicomodule of the *same variance* as B. There is a canonical such reading, and it sits visibly inside B.

Definition 5.1 (constant-branch slice)**.** Writing a position of B as (s, g) with $s \in S$, $g \in T^S$, let

$$B' \;:=\; \{(s,g) \in B :\; g \in \mathrm{const}(T)\}, \qquad \mathrm{const}\colon T \hookrightarrow T^S,\; t \mapsto \mathrm{const}_t,$$

with directions inherited ($S \times T$ per position). As a polynomial, B' has positions $S \times \mathrm{const}(T) \cong S \times T$ and directions $S \times T$, i.e. $B' = (S \times T) \cdot \mathrm{y}^{S \times T} = \Delta(S \times T) = A$ on the nose.

Lemma 5.2 (B' is a sub-bicomodule)**.** *B' is closed under the weld coactions $\lambda = \delta_{\Delta S} \triangleleft \Delta T$ and $\rho = \Delta S \triangleleft \delta_{\Delta T}$; hence $(B', \lambda|, \rho|)$ is a $(\Delta T, \Delta S)$-sub-bicomodule of B.*

Proof. Both coactions leave the branch-map coordinate $g = (t_s)_{s \in S}$ *pointwise* unchanged, only reorganising positions. For $\rho = \Delta S \triangleleft \delta_{\Delta T}$: the inner $\delta_T(t, \psi) = (t,\, j \mapsto (j, \psi))$ keeps the first coordinate t, so the B-layer of $\rho(s, g)$ has the same g; a constant g stays constant. For $\lambda = \delta_{\Delta S}$: $\delta_S(s, \varphi) = (s,\, i \mapsto (i, \varphi))$ duplicates the whole ΔT-ground with the same φ, hence the same g; constant stays constant. The counit and (co)associativity laws restrict because they are the comonoid laws of $\Delta S, \Delta T$ whiskered, and B' is full on positions with inherited directions. Machine-checked (ρ-closed, λ-closed) at $(|S|, |T|) \in \{(2,2), (2,3), (3,2), (1,2), (2,1)\}$. □

Proposition 5.3 (the worker's canonical reading)**.** *The map $\sigma\colon A \to B$ with $\sigma_1(s,t) = (s, \mathrm{const}_t)$ on positions and $\sigma^\sharp = \mathrm{id}_{S \times T}$ on directions is an isomorphism of $(\Delta T, \Delta S)$-bicomodules $A \cong B'$. Equivalently, A carries the $(\Delta T, \Delta S)$-bicomodule structure obtained by cobase change of the regular (A, A)-bicomodule along the comonoid projections $\pi_S\colon A \to \Delta S$ (left) and $\pi_T\colon A \to \Delta T$ (right) — the indiscrete functors induced by the set projections — and this structure coincides with B'.*

Proof. σ_1 is the bijection $S \times T \cong S \times \mathrm{const}(T) = B'$-positions, and $\sigma^\sharp = \mathrm{id}_{S \times T}$ matches directions. By Lemma 5.2 the coactions act on B' exactly as they do on A under this identification (both leave the g-value alone), so σ commutes with λ, ρ and is a bicomodule isomorphism onto its image B'. The sub-bicomodule description is primary and fully verified; the cobase-change description is the conceptual justification that B' is the *canonical* reading (not an arbitrary sub-object), and agrees with it on carriers and positions. □

Theorem 5.4 (worker versus weld: a Σ-index separation)**.** *Read A and B as $(\Delta T, \Delta S)$-bicomodules via Proposition 5.3 ($A \cong B' \subseteq B$). Then*

$$\bar{F}_B(V) = \sum_{g \in T^S} V^S = T^S \cdot V^S, \qquad \bar{F}_A(V) = \sum_{g \in \mathrm{const}(T)} V^S = T \cdot V^S.$$

Worker and weld have the same *content power $(-)^S$ (directions S); they differ* only *in the Σ-aggregation index, T^S (all branch maps) for the weld versus $T \cong \mathrm{const}(T)$ (constant branch maps) for the worker. Hence $\bar{F}_A$ is a sub-prafunctor of $\bar{F}_B$, with $\bar{F}_A \cong \bar{F}_B$ iff $|S| = 1$ or $|T| = 1$, and a* proper *inclusion exactly when $|S| \geq 2$ and $|T| \geq 2$.*

Proof. $\bar{F}_B = \sum_{g \in T^S} V^S$ is Theorem 4.4; its Σ-index is the set of B-positions modulo the left ΔS-collapse, namely T^S (the S-head is killed by indiscreteness), and the content exponent is the surviving direction S (the T-direction is killed by the right ΔT-collapse). By Proposition 5.3, $A \cong B'$ is B restricted on positions to $\mathrm{const}(T) \subseteq T^S$ with directions unchanged; the same recipe

restricts the Σ-index to $\mathrm{const}(T)$ and leaves the content power $(-)^S$ untouched, giving $\bar{F}_A(V) = \sum_{t\in T} V^S = T \cdot V^S$. Equality of the two functors for all V is equality of polynomials $T^S\mathsf{y}^S$ and $T\mathsf{y}^S$, i.e. $|T^S| = |T|$, i.e. $m^n = m$ (with $n = |S|$, $m = |T| \geq 1$), i.e. $n = 1$ or $m = 1$. Otherwise $T \subsetneq T^S$ and the inclusion $\bar{F}_A \hookrightarrow \bar{F}_B$ is proper. □

Remark 5.5 (what this corrects)**.** The content *exponents agree* — both are $(-)^S$. An earlier, superseded argument read A *without* the right ΔT-coaction and so reported content exponent S for the weld against $S \times T$ for the worker, concluding a mismatch already at $|S| = 1$. But at $|S| = 1$ one has $T^S = T$ and $B = \mathsf{y} \triangleleft \Delta T = \Delta T = \Delta(1 \times T) = A$ on the nose: worker = weld there, so the mismatch claim was an artefact of the inconsistent reading. With the consistent $(\Delta T, \Delta S)$-reading the T-directions of A collapse into the exponent S, and the true separation is in the Σ-index, T versus T^S, nonzero exactly where non-constant branch maps exist: $|S| \geq 2$ and $|T| \geq 2$.

What the retract forgets. The states-entanglement discipline comes with a retraction $r\colon B \to A$, $r_1(s, g) = (s, g(s))$ (self-evaluation), splitting σ; the idempotent $e = \sigma \circ r$ has $e(s, g) = (s, \mathrm{const}_{g(s)})$.

Theorem 5.6 (the retract's keep/discard)**.** *The idempotent $e\colon S \times T^S \to S \times T^S$ has image $S \times \mathrm{const}(T) \cong S \times T$ (the worker positions). For each current input s, e keeps the value $g(s) \in T$ and discards the restriction $g|_{S\setminus\{s\}} \in T^{S\setminus\{s\}}$, a fibre of size $|T|^{|S|-1}$. Concretely the fibre of e over (s, const_t) is $\{g : g(s) = t\} \cong T^{S\setminus\{s\}}$.*

Proof. $e \circ e = e$ since $\mathrm{const}_{g(s)}$ evaluated at s is $g(s)$; the image is the constant slice. Two maps g, g' satisfy $e(s, g) = e(s, g')$ iff $g(s) = g'(s)$; fixing $g(s) = t$ leaves $S \setminus \{s\}$ free, so the fibre is $T^{S\setminus\{s\}}$. Idempotency, image, and all fibre sizes machine-checked at the five shapes above. □

Remark 5.7 (the Σ-tail, stated honestly)**.** The weld's Σ-aggregation index is the *full* branch-map set $T^S = \mathrm{const}(T) \sqcup \mathrm{NC}$, with $|\,\mathrm{const}(T)| = m$ and $|\mathrm{NC}| = m^n - m$ (non-constant maps). It is *not* correct to say "T^S = the non-constant maps" (a cardinality contradiction: $m^n \neq m^n - m$), nor that the retract forgets exactly the non-constant maps: per input s the retract forgets the off-diagonal fibre $T^{S\setminus\{s\}}$ of size m^{n-1}, which is neither NC ($m^n - m$) nor T^S (m^n). At $(|S|, |T|) = (2, 3)$ these are 3 versus 6 versus 9, all distinct. What is correct: the weld's Σ-tail *is* the full branch-map set T^S, and the retract realises its projection onto the worker's constant slice $\mathrm{const}(T) \cong T$ by "evaluate at the current input, then make constant."

Remark 5.8 (no reversal of adjointness)**.** The superseded draft worried that the retract discards "exponential, hence right-adjoint-flavoured" data T^S while the SGF factorization discards a coproduct. There is no paradox. T^S appears as the *indexing set* of the left-adjoint coproduct $\Sigma_{T^S} \dashv \Delta_{T^S}$, which is a left adjoint for any index set, exponential or not. The genuinely right-adjoint (connected-limit) content $(-)^S$ is *kept* in the $F/1$ factor, exactly as [4] predicts, and is shared by worker and weld.

Resolution of the former Conjecture 5.1. The superseded 2026-09-20 draft conjectured that "a worker is a parametric right adjoint / bicomodule, and the retract kernel is the aggregation tail." We can now decide it precisely.

Theorem 5.9 (decision of the former conjecture)**.** *Let S, T be nonempty.*

1. ***(existence — true, peer-reviewed.)*** *The weld $B = \Delta S \triangleleft \Delta T$ is a bicomodule: the regular $(\Delta T, \Delta S)$-bicomodule (Theorem 4.2), representing the prafunctor $\bar{F}_B(V) = (T \times V)^S$ (Theorem 4.4).*
2. ***(identification — corrected.)*** *The worker A, read as the $(\Delta T, \Delta S)$-bicomodule $\sigma(A) = B' \subseteq B$, is* not *the full weld-prafunctor but its* constant-branch sub-bicomodule*. Worker and weld share the right-adjoint content power $(-)^S$ and differ only in the Σ-index (T versus T^S). The inclusion is proper exactly on $|S| \geq 2 \wedge |T| \geq 2$ (witness $(2, 2)$ at $V = 2$: $\bar{F}_A = 8$, $\bar{F}_B = 16$); worker = weld on the nose when $|S| = 1$ or $|T| = 1$.*

(3) ***(tail — true.)*** *The weld's Σ-aggregation tail is the coproduct over the branch-map set $T^S = \mathrm{Hom}(S,T)$; the retract projects this index onto the worker's constant slice (Remark 5.7).*

In particular the former conjecture holds in the corrected form: the worker is a parametric right adjoint — the constant-branch sub-bicomodule of the weld — not the whole weld-prafunctor.

Remark 5.10 (trust)**.** Clause (1) (Theorems 4.2, 4.4) is *peer-reviewed* (Rick). The worker-as-constant-branch-slice content of clauses (2), (3) (Definition 5.1, Lemma 5.2, Propositions 5.3, Theorems 5.4, 5.6) is *proved* and machine-checked at the five shapes $(|S|,|T|) \in \{(2,2),(2,3),(3,2),(1,2),(2,1)\}$; the full coaction-by-coaction identification of the two bicomodule structures of Proposition 5.3 for general S,T is not written out here, as the comparison only uses $A \cong B'$.

6. What this buys the program

The weld now has a precise home in **Poly**: it *is* a data-migration prafunctor $\Delta T\text{-}\mathbf{Set} \to \Delta S\text{-}\mathbf{Set}$, $X \mapsto (T \times X)^S$ — "read the whole input state S; return, for each input, an output token in T and a value in X," with the branch-map choice aggregated over T^S by Σ. This places the *state* axis of the compositional-correctness program inside the very framed bicategory $\mathbf{Cat}^\sharp$ that already houses the directed (retrofunctor) axis.

But the identification "worker = parametric right adjoint content" is *false* on $|S|,|T| \geq 2$: state-entanglement (the worker A) and data-aggregation (the weld's Σ-tail) are genuinely different collapses of the same content power $(-)^S$. The worker keeps a branch choice per output token $t \in T$; the weld keeps a branch choice per *function* $g \in T^S$. Keeping the two distinct is a semantic requirement of agent-orchestration, not a bug — and the exact locus where it matters is $|S|,|T| \geq 2$. That is the honest, sharpened dividend of running the SGF correspondence on this one bicomodule.

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